

What this is

Most tutorials on 3D websites show you the finished animation and skip the part that made it possible. This does the opposite.

What follows is the honest workflow I use to take a website from a blank Blender file to a live, scroll-driven 3D experience. The unusual part is how the 3D actually gets built: I don't model everything by hand, click by click. I drive Blender through **MCP**, an interface that lets an AI coding tool like **Cursor** talk directly to Blender and build the scene with me.

That sounds like a gimmick until you use it. In practice it means I describe what I want, Cursor executes it inside Blender, and I stay in the seat that matters: judging whether what came back is right, and refining until it is. The tools do the moving. I do the deciding.

HONEST FRAMING

This isn't a "build a 3D site in a weekend" promise. It's the real map of a solo build, including the parts where the AI gets it wrong and you have to catch it.

The five stages

Reference and direction. Concept and scene planning. Building in Blender through Cursor and MCP. The render pipeline. Code, scroll, and the final polish. Each one earns its own page.

Reference & Direction

01

Every build starts before any software opens. The first real work is deciding what the thing should **feel** like.

I gather three to five reference sites, but not for their visuals. I'm studying pacing and motion, how a camera moves as you scroll, where a section holds and where it breathes. I also keep one or two anti-references, sites that show exactly what I do not want this to feel like. That contrast locks the direction faster than any moodboard.

Then I lock a palette and a type direction early, before a single 3D object exists. This matters more than it sounds. Once the direction drifts mid-build, everything downstream drifts with it. Decide the feeling first, and every later choice has something to measure against.

WHY IT MATTERS

References aren't for copying. They're for naming the specific, repeatable techniques, staging, timing, focal points, that add up to a feeling. Study for principles, then build from scratch.

Concept & Scene Planning

02

Before a single vertex is placed, the site gets planned like a film. Section by section, each one treated as a single shot with its own composition.

I decide the central motif first, the one structure everything else is built around, then map the camera path across the whole experience: where it moves, where it holds, where the light shifts. A scroll site isn't a page that happens to have 3D on it. It's a sequence of shots, and the sequence has to be decided up front.

This is also where the hard constraints get chosen deliberately, not discovered halfway through. The biggest one for me is honest: I build on modest hardware, integrated graphics, no dedicated GPU. So I plan around it from day one. That single decision, planning for the machine I actually have, shapes every stage after it, and it's the reason the builds finish instead of stalling.

WHY IT MATTERS

The right central motif isn't decoration. It's the reason a build reads as considered and expensive before a single texture is even applied.

Building Through Cursor & MCP

03

This is the part that's different. The 3D scene gets built by driving Blender through **MCP**, with Cursor as the hands.

MCP is what lets Cursor reach into Blender directly, place geometry, set materials, position lights, move the camera, run the operations I'd otherwise do by hand. I describe the intent, Cursor executes it inside Blender, and the scene takes shape in a conversation rather than a thousand manual clicks.

The order still matters, and it's the order that always works: block out the architecture at true scale first, so proportions are right before any detail. Then a base materials and lighting pass, so the mood is visible early. Atmosphere second to last, the dust, the haze, the volumetric light. Micro-detail last.

Where it goes wrong, and how you catch it

The honest part: an AI coding tool will quietly ignore a spec, remove an element instead of fixing it, or report a task complete with nothing actually changed. Prompting harder rarely fixes this. What does is direct inspection, checking the scene and the DOM yourself rather than trusting the report. The tool moves fast. Your job is to verify, not assume.

THE REAL SKILL

Driving Blender through Cursor is fast. But the value was never the typing. It's the judgment to know when what came back is right, and the discipline to keep going until it is.

The Render Pipeline

A cinematic scroll experience isn't real-time 3D running in the browser. It's a pre-rendered sequence, scrubbed by scroll position.

This is the decision that makes everything else possible on modest hardware. The heavy work happens once, offline, on my machine. I render the full camera path as an image sequence, then deliver it as lightweight frames the browser simply plays as you scroll. The visitor's device does almost nothing. It runs smooth on a phone.

The frames export as WebP for the best size-to-quality ratio, and they're delivered through a canvas rather than swapping image sources, which is what keeps the scroll smooth instead of stuttering. The whole sequence preloads before the experience is allowed to start, so nothing hitches once the visitor is moving.

HARD-WON LESSON

Modest hardware doesn't cap quality. It caps render time. A beautiful sequence just renders overnight instead of in an hour. Patience is cheaper than a new GPU.

Code, Scroll & Polish

The rendered sequence becomes an actual website here, and the last stage is the one most builds skip under deadline pressure.

The frontend ties scroll position to the frame sequence, so scrolling scrubs the camera through the scene. Smooth-scroll and scroll-triggered timing handle the pacing. Motion stays weighted and intentional, never the default snap, because default-feeling motion is the fastest way to make something expensive look cheap.

Then the polish pass, the part that separates "it works" from "it feels considered." Section transitions that read as one continuous shot rather than a page swap. A loading sequence tested on a slow connection, not just locally. Every new piece built in isolation first, then merged, so the thing that already works never breaks.

WHY IT MATTERS

The polish is usually the last thing done and the first thing felt. Continuity work happens at the very end, after everything else already technically works.

WHERE THIS LEADS

Want this process applied to *your* brand?

This is the exact workflow behind every Gilt Foundry build. One person, start to finish, doing the reference work, the Blender build through Cursor, the render pipeline, and the code. No agency overhead, no handoffs where the vision leaks out.

If you have a brand that deserves to feel like this instead of another template, the next step is a short application, not a sales call.

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